

Chapter 1B: Intelligent Agents

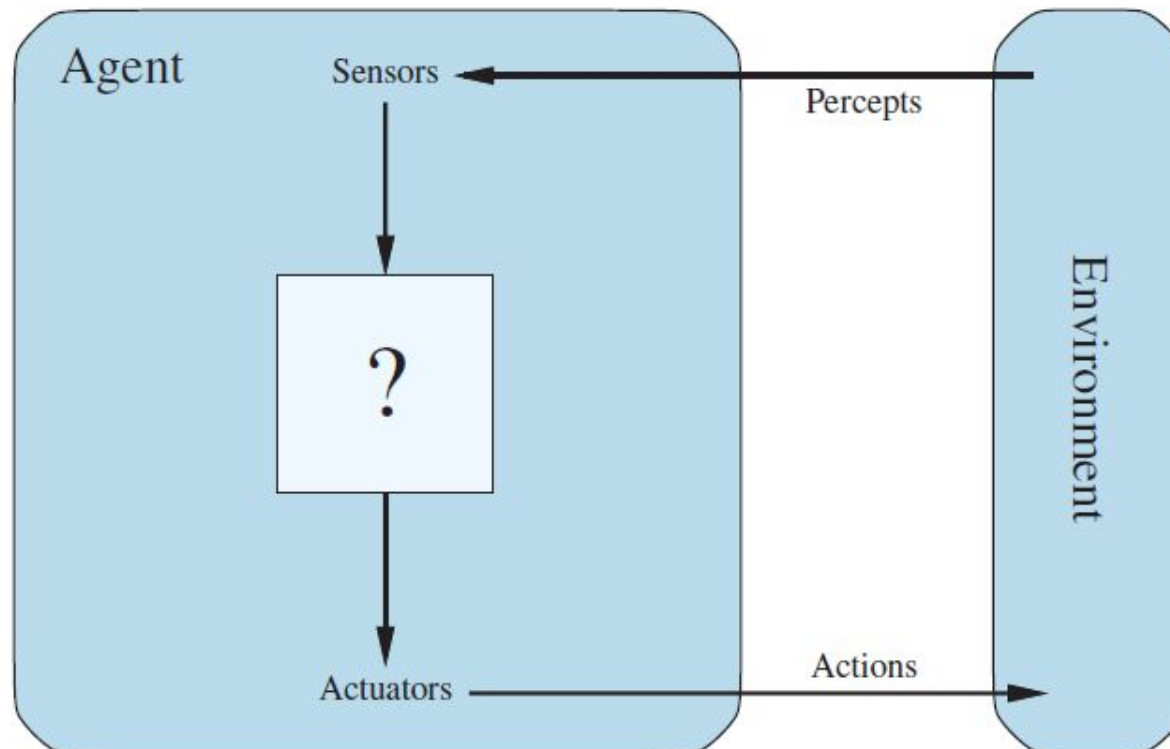
AMCS2104 Fundamentals of Artificial Intelligence

Content

- Agents
- Agents and Environments
- Agent Functions
- The Concept of Rationality
- Performance Measure
- Rational Agents
- Omniscience
- Information Gathering, Autonomy and Learning

Agents

- An **agent** is anything that can be viewed as perceiving its **environment** through **sensors** and acting upon that environment through **actuators**.



Agents: Examples

□ Human agent

- **Sensors:** Eyes, ears, and other organs.
- **Actuators:** Hands, legs, vocal tract, etc.
- **Environment:** ?

□ Robotic agent

- **Sensors:** Cameras and infrared range finders.
- **Actuators:** Various motors and actuators.
- **Environment:** ?



Agents: Examples

□ Software agent

- **Sensors:** Keyboard, mouse, touchscreen and microphone.
- **Actuators:** Display screen and speaker.
- **Environment:** ?



Agents and Environments

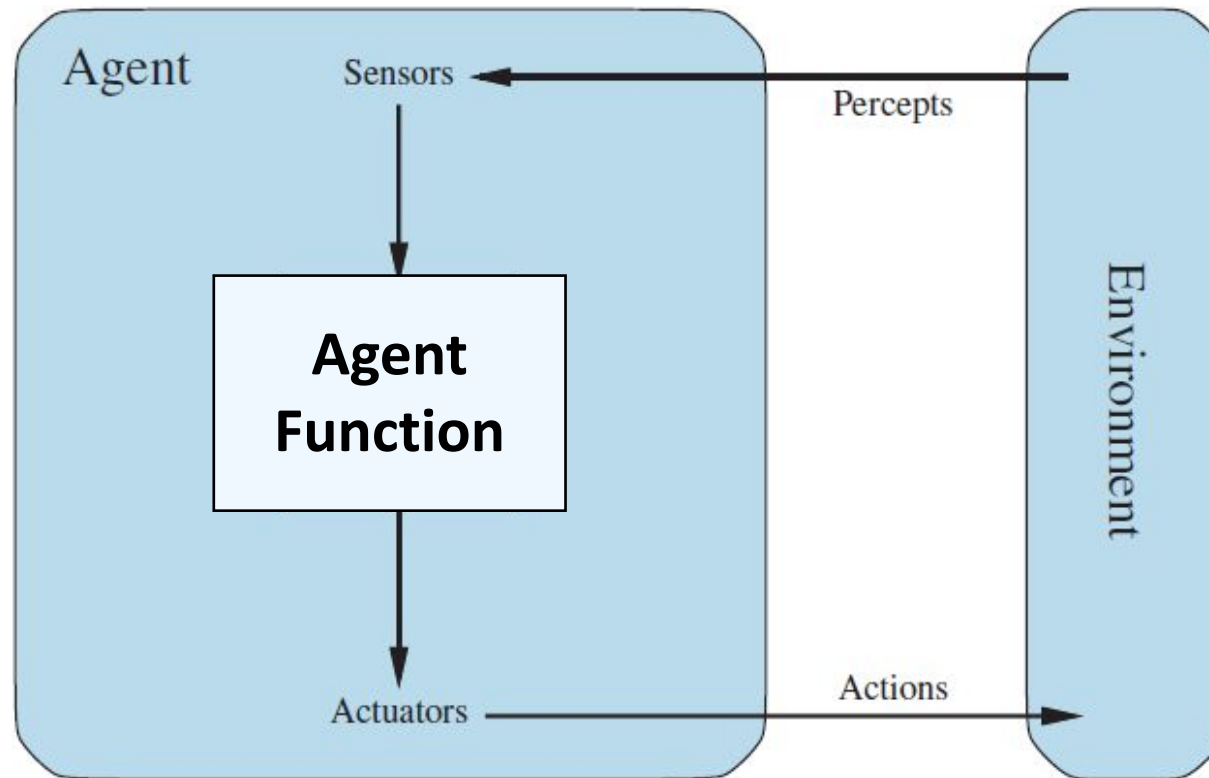
- Agents use their **sensors** to capture information from the **environment** setup and then process the data for further **actions**.
- Agents **interact** with the **environment** in 2 ways:
 - **Perception**
 - A **passive interaction** where it gains information about the environment without changing it.
 - **Action**
 - An **active interaction** where the environment is changed after the action is performed.

Agents and Environments

- An agent's **choice of action** at any given instant can depend on its **built-in knowledge** and the **entire percept sequence observed to date**, but not on anything it hasn't perceived:
 - **Percept** refers to the agent's perceptual inputs at any given instant.
 - **Percept sequence** is the complete history of everything the agent has ever perceived.

Agent Function

- An **agent's behavior** is described by the **agent function** that **maps any given percept sequence to an action**.

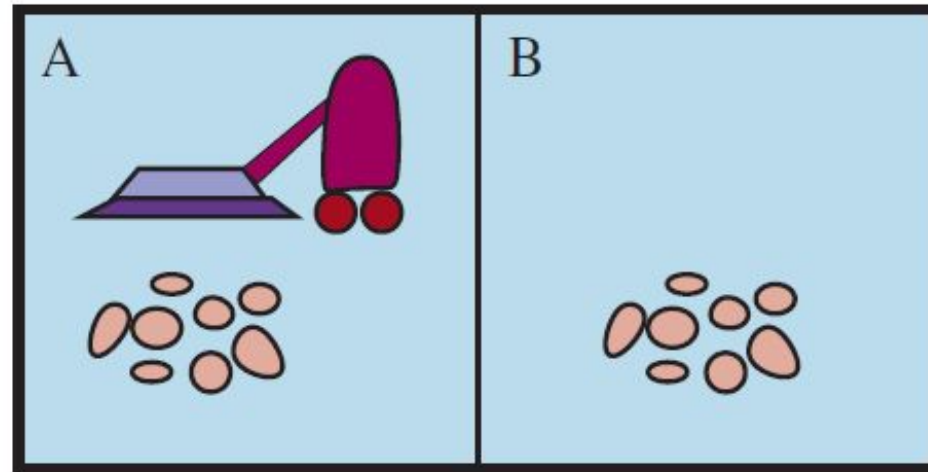


How Agent Function Works?

- An example of how an agent function can be tabulated

Percept Sequence	Action
Percept Sequence #1	Action #1
Percept Sequence #2	Action #2
Percept Sequence #3	Action #3
...	...
Percept Sequence #($n - 1$)	Action #($n - 1$)
Percept Sequence # n	Action # n

The Vacuum World: Agent Function



- **The vacuum world** is a made up world where:
 - Everything that happens can be described.
 - Many variations can be invented.

The Vacuum World: Agent Function

- The vacuum world has two locations only:
 - Square A and square B.
- The vacuum agent **perceives**:
 - Which square it is in.
 - Whether there is dirt in the square.
- The agent can perform the following **actions**:
 - Left
 - Right
 - Suck

The Vacuum World: Agent Function

□ A partial tabulation of this **agent function** is shown in the following table:

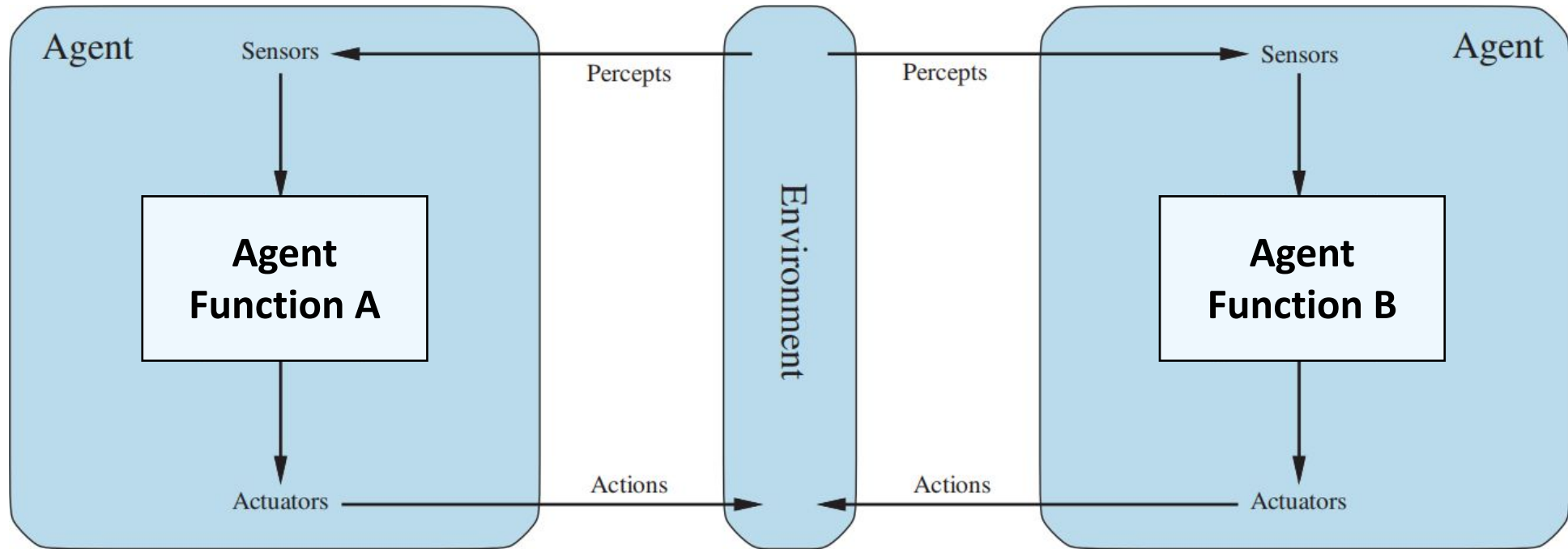
Percept sequence	Action
<i>[A, Clean]</i>	<i>Right</i>
<i>[A, Dirty]</i>	<i>Suck</i>
<i>[B, Clean]</i>	<i>Left</i>
<i>[B, Dirty]</i>	<i>Suck</i>
<i>[A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>
<i>[A, Clean], [A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>

How Agent Function Works?

- The table is only an **external characterization** of the agent; internally, the **agent function** will be implemented by an **agent program**.
- **Agent function** is an **abstract mathematical description**.
- **Agent program** is a **concrete implementation**, running within some physical system (see a sample implementation provided below).

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
    if status = Dirty then return Suck
    else if location = A then return Right
    else if location = B then return Left
```

The Concept of Rationality



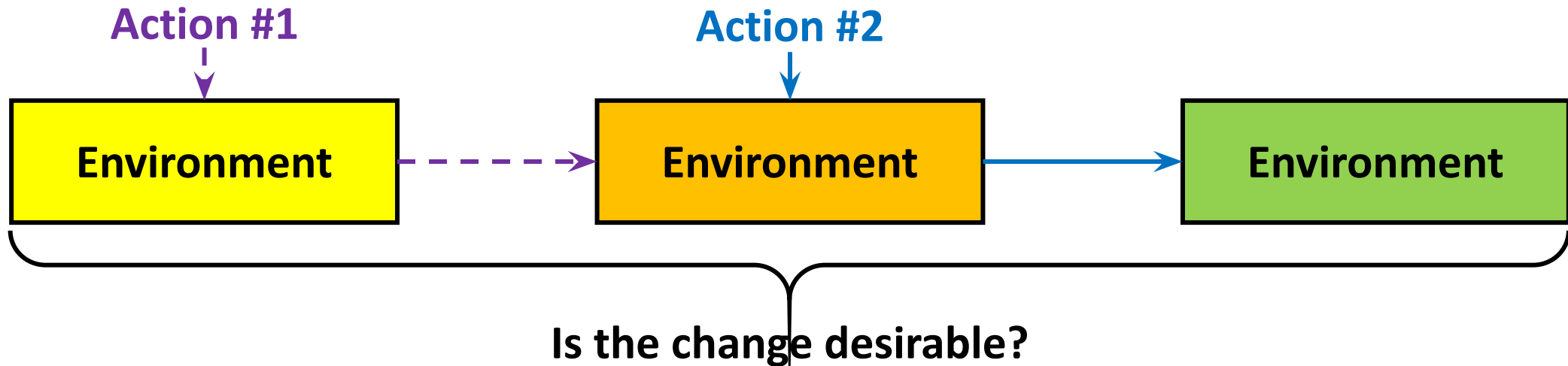
- A **rational agent** is one that **does the right thing**.
- Obviously, doing the right thing is better than doing the wrong thing, but what does it mean to do the right thing?

The Concept of Rationality

- In the context of A.I., **consequentialism** is a theory used to determine **what is right and wrong**.
- The consequentialist believes that **outcomes are all that matter**.
- In other words, when an action leads to a **desired result**, then it is considered the right thing to do.
- Unlike humans, however, machines do not have desires and preferences of their own.
- Therefore, **performance measure** is designed to help machines capture the **notion of desirability**.

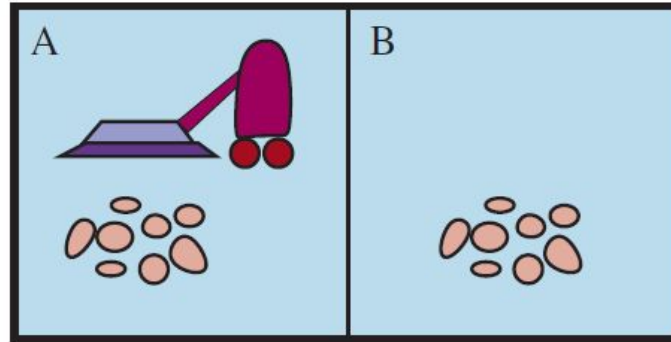
Performance Measure

- An agent's **actions** will **change the environment** by causing it to **go through a sequence of states**.



- This notion of **desirability** is captured by a **performance measure** that evaluates any given sequence of **environment states**.

The Vacuum World: Example #1



□ Performance measure

□ **Performance:** Based on the amount of dirt cleaned up in a single 8-hour shift.

□ **Good performance:** A higher amount of dirt is cleaned up.

□ **Bad performance:** A lower amount of dirt is cleaned up.

□ **Question:** How would an agent maximize its performance?

The Vacuum World: Example #2

- **Performance measure**

- **Performance:** One point will be awarded for each clean square at each time step.

- **Good performance:** A higher total score.

- **Bad performance:** A lower total score.

- **Question:** How would an agent maximize its performance?

The Vacuum World: Example #3

- **Performance measure**

- **Performance:**

- Based on the cleanliness of the floor where:

- One point will be awarded for each clean square at each time step.

- One point will be deducted for electricity consumed and noise generated.

- **Good performance:** A higher total score.

- **Bad performance:** A lower total score.

- **Question:** How would an agent maximize its performance?

Choosing the Right Performance Measure

□ Conclusion

- With a "rational" agent, of course, **what you ask for is what you get.**

□ Rule of Thumb

- Design **performance measures** according to **what one actually wants to be achieved in the environment**, not according to how one thinks the agent should behave.

Rationality

- At any given time, **rationality** depends on 4 things:
 - The **performance measure** that defines the criterion of success.
 - The **agent's prior knowledge** of the environment.
 - The **actions** that the agent can perform.
 - The agent's **percept sequence to date**.
- **Definition of a rational agent**
 - For each possible **percept sequence**, a rational agent should select an **action** that is expected to **maximize its performance measure**, based on the **evidence** provided by the **percept sequence** and its **built-in knowledge**.

Rationality: Example

□ **Question:** Is an agent with this agent function considered as a rational agent?

Percept sequence	Action
<i>[A, Clean]</i>	<i>Right</i>
<i>[A, Dirty]</i>	<i>Suck</i>
<i>[B, Clean]</i>	<i>Left</i>
<i>[B, Dirty]</i>	<i>Suck</i>
<i>[A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>
<i>[A, Clean], [A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>

Rationality: Example #1

□ Performance measure

- Awards one point for each clean square at each time step, over a "lifetime" of 1000 time steps.

□ Prior knowledge

- The "**geography**" of the environment is **known a priori**.
- The dirt distribution and the initial location of the agent are not known.
- "**Suck**" cleans the current square, and clean squares stay clean.
- The "**Left**" and "**Right**" actions move the agent to the left and right but not to the outside of the environment.

Rationality: Example #1

□ Perception

- Correctly perceives its location (Square A or B).
- Correctly perceives whether that location contains dirt.

□ Actions

- Left , Right, and Suck.

- Under **these circumstances** the agent is indeed rational because its **expected performance** is at least as high as any other agent's.

Rationality: Example #2

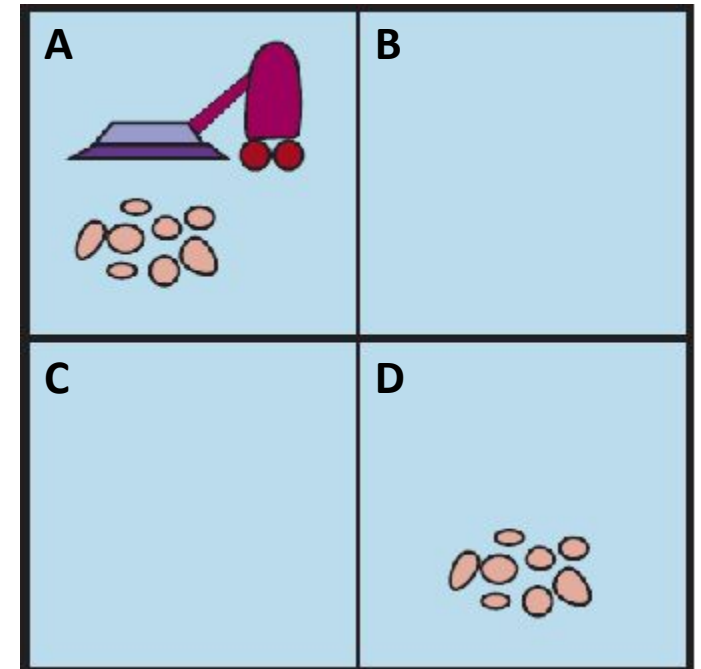
- **Question:** Will the agent still be considered as a rational agent if the **performance measure is changed** as follows while others remain the same?
- **Performance measure:**
 - Award one point for each clean square at each time step, over a "lifetime" of 1000 time steps.
 - Deduct one point for each movement to the left or right.

Rationality: Example #3

□ **Question:** Will the agent still be considered as a rational agent if it has **incomplete prior knowledge** as follows while others remain the same?

□ **Prior knowledge:**

- The agent lives in a world with 2x2 4-square world but it "thoughts" that it is living in a 2-square world.
- The rest remains the same.

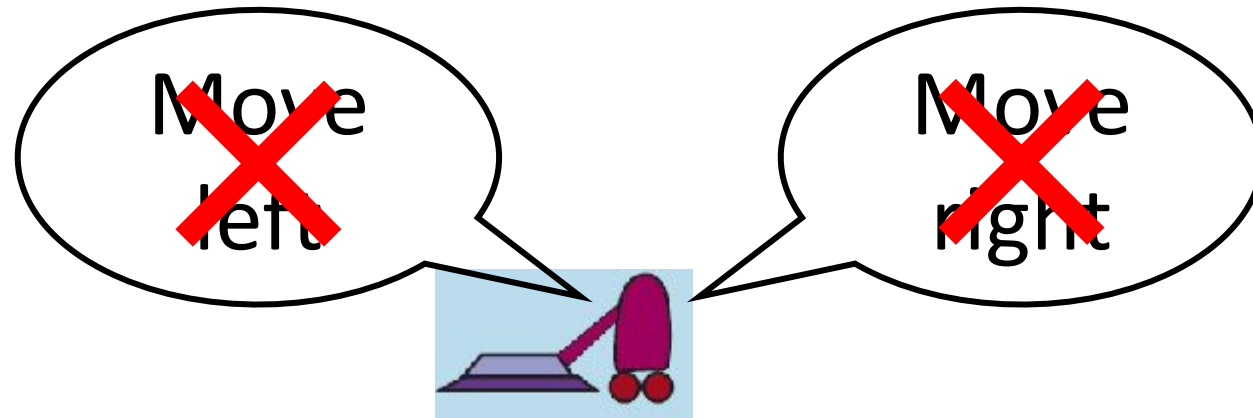


Rationality: Example #4

□ **Question:** Will the agent still be considered as a rational agent if it is **not capable of movement** while others remain the same?

□ **Actions:**

□ Suck only.



Rationality: Example #5

□ **Question:** Will the agent still be considered as a rational agent if **both of its location and dirt sensors malfunction** while others remain the same?

□ **Percept sequence:**

□ Not up-to-date.



Rationality vs. Omniscience

- Omniscience carries the meaning of "the state of knowing everything".
- Omniscient agents **know the actual outcome of their actions** and **can act accordingly**.



Movies: "Dr. Strange" (2016)



Movies: "Avengers: Infinity War" (2018)

Rationality vs. Omniscience

□ Rational agent

- Doesn't know the actual outcome of its actions.
- Acts under **uncertainty** most of the time.
- Aims to maximize its performance by achieving the **best expected outcome**.

□ Omniscient agent

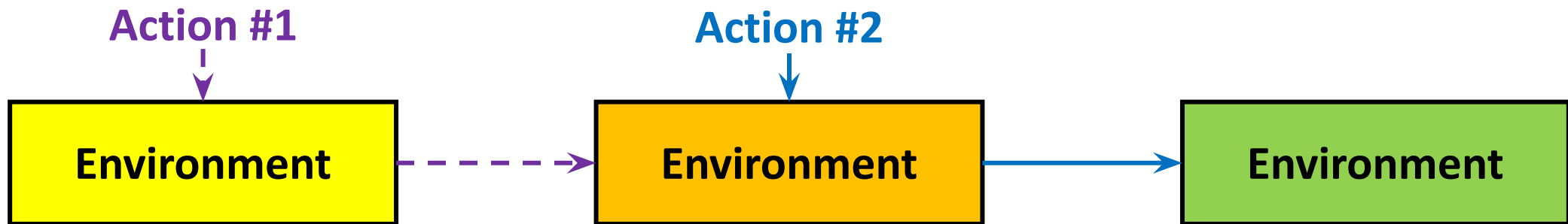
- Knows the actual outcome of its actions.
- Acts under **certainty** all of the time.
- Aims to maximize its performance by achieving the **best actual outcome**.

Rationality

- Our definition of **rationality** does not require omniscience because a rational choice depends only on **the percept sequence to date**.
- However, a rational agent should possess the following 3 capabilities:
 - **Information gathering**
 - **Autonomy**
 - **Learning**

Rationality: Information Gathering

- A rational agent should be capable of **information gathering**.
 - Perceive the environment before making any action.
 - Act on the environment to change it can affect the future percepts.



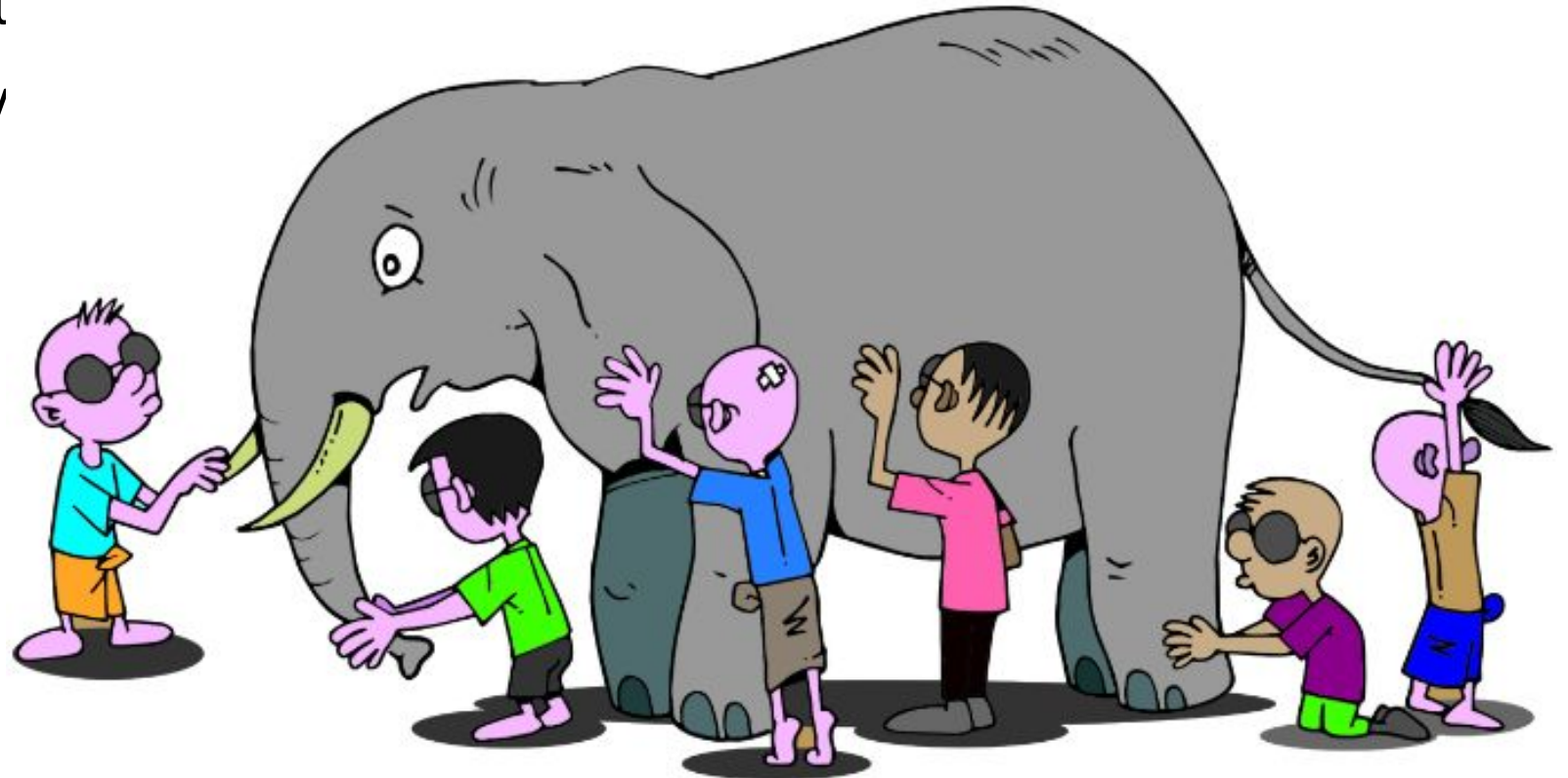
- Being able to do so can help maximize an agent's expected performance.

Rationality: Learning

- Agents that are capable of learning will **learn as much as possible from what they perceive**.
- Since no designers are omniscient agents, most of the environments are said to be not completely known a priori.
- The **learning capability** can compensate for agents' **incomplete** or **incorrect** prior knowledge about the environment, making them more **adaptable** to the environments.

Rationality: Learning

- Therefore, the incorporation of **learning** allows one to design a rational agents that can:
 - Demonstrate a better performance
 - Succeed in a vast variety of environments.



Rationality: Autonomy

- ❑ An agent **lacks autonomy** if it solely relies on the prior knowledge of its designer rather than on its own percepts and learning processes.
- ❑ Agents that over-rely on the prior knowledge are fragile and lack **autonomy**.
- ❑ A rational agent should be **autonomous**, in which it can act based on its own experience in to achieve specific goals, without being directed to do so.



Prior Knowledge vs. Experience

- **Question:** Which one is more important, the prior knowledge of the environment or the experience gained by oneself?
- An agent seldom requires complete autonomy from the start.
- Provide an agent with some initial knowledge as well as the ability to learn.
- After sufficient experience of its environment, the behavior of a rational agent can become effectively independent of its prior knowledge.

Reading Homework

- **Artificial Intelligence: A Modern Approach**, 4th Edition, by Stuart Russell and Peter Norvig
 - Chapter 2: Intelligent Agents (page 54 to page 61)

The End